

InspectAgent: testing instructions

Code: <https://github.com/najmunnaharhira/OpenCV-AI-Competition-2026-powered-by-AWS>

1. Use the live demo (recommended)

1. Open the demo URL listed in the submission's "Try it out" links.
2. Pick a part from the drop-down. Each entry shows the capture problem (blur, dark, bright, glare, none) and the ground-truth defect.
3. Press **Inspect**. The page shows the decision, the agent trace and the evidence images.
4. Things to try:
 - A part with capture **blur**: the trace shows the Laplacian sharpness below 120, the agent recaptures with refocus, then aligns, detects and decides.
 - A part with truth **faint_spot**: no defect at the production threshold, but the residual peak triggers `inspect_closeup`, which finds the spot and rejects the part.
 - A part with truth **missing_hole**: auto-reject is blocked by a guardrail. The agent requests human approval to quarantine the lot. Press **Approve** or **Decline** to record the decision.
 - Switch the planner to **Claude on Bedrock** to see the same tools driven by Claude.

2. Run locally (no AWS account needed)

Requires Python 3.12 (3.10+ works).

```
git clone https://github.com/najmunnaharhira/OpenCV-AI-Competition-2026-powered-by-AWS
cd OpenCV-AI-Competition-2026-powered-by-AWS
python3 -m venv .venv && . .venv/bin/activate
pip install -r requirements-dev.txt
pytest                                # 12 tests, about 10 s
python -m eval.run_eval --n 200 --seed 11 # agent vs. static baseline table
uvicorn inspectagent.api:app            # then open http://127.0.0.1:8000
```

GET /healthz reports the OpenCV version (5.0.0) and the CPU architecture.

3. Run the Claude planner on Amazon Bedrock

Enable Claude model access in the Bedrock console, then:

```
export AWS_REGION=us-east-1 BEDROCK_MODEL_ID=anthropic.claude-opus-5-5
python -m eval.run_eval --n 40 --planner claude
```

If Bedrock is unreachable or the model refuses, the loop falls back to the rule planner and the trace records a `planner_error` event.

4. Deploy your own copy to AWS (Graviton)

```
cd deploy
sam build --use-container
sam deploy --guided          # Architecture=arm64 by default
```

The DemoUrl stack output is the endpoint. Traces and evidence go to S3, and approval requests go to DynamoDB with an SNS notification.

5. Expected results (seed 11, 200 synthetic parts)

metric	static baseline	agent
pass/fail accuracy	0.720	0.990
false accept rate	0.160	0.020
false reject rate	0.400	0.000